# SMART WOMEN SAFETY WEARABLE

A Low-Cost Emergency Alert and Location Tracking System | AURA 2K26

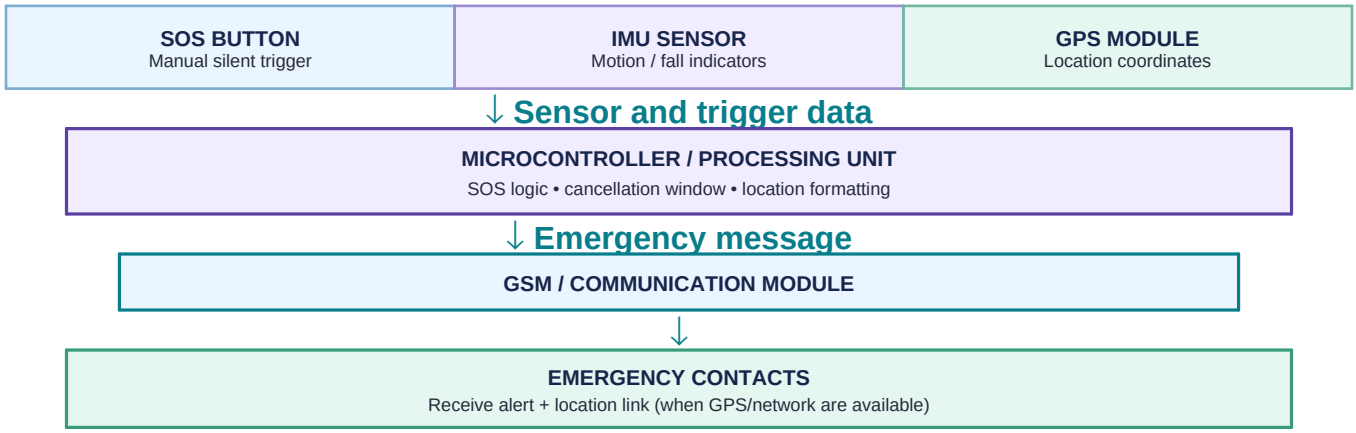
## 1. PROBLEM STATEMENT

During an emergency, a person may not be able to unlock a phone, make a call, or openly ask for help. Basic safety devices may rely only on a manually activated alarm and may not share location information or help limit accidental alerts. This project proposes a compact wearable that can discreetly notify trusted contacts and share the user's location when available.

## 2. PROPOSED METHODOLOGY

The prototype is planned around a microcontroller, SOS push-button, inertial measurement unit (IMU), GPS module, and a suitable communication module such as GSM. It may be worn as a bracelet, pendant, or keychain. A long press on the SOS button will initiate Silent SOS. If GPS data is available, the system will prepare a location link and send an alert to predefined emergency contacts. The IMU will be evaluated for sudden falls or unusual movement patterns; these are possible emergency indicators, not proof of an assault. A short cancellation window is proposed to reduce false alerts. Evidence logging and a monitoring dashboard may be future extensions.

## 3. PROPOSED BLOCK DIAGRAM




## 4. INNOVATION & PROPOSED OUTCOME

The intended contribution is to combine discreet manual SOS, location sharing, and exploratory movement-based alerting in one portable device, with a cancellation step to reduce accidental notifications. The design prioritizes accessible components and a cost-conscious, modular build rather than an expensive AI computer. The prototype is under development. Controlled tests will evaluate SOS activation, GPS availability and accuracy, message delivery, battery/runtime, and false-trigger handling. The expected outcome is a proof-of-concept that can alert trusted contacts and provide useful location information when GPS and network coverage are available.